

Final Report: Evaluation of GO:0004721 (Phosphoprotein Phosphatase Activity) for *Pseudomonas putida* AceK (Q88EA1)

Executive Judgment

Verdict: Partially supported / too general — Failure mode #1 (granularity / family-vs-subfamily).

The seed hypothesis that AceK (Q88EA1) from *Pseudomonas putida* KT2440 possesses phosphoprotein phosphatase activity (GO:0004721) is **correct in substance but insufficiently specific**. AceK is a genuine bifunctional isocitrate dehydrogenase kinase/phosphatase whose phosphatase catalytic residues are fully conserved relative to the experimentally characterized *E. coli* ortholog (P11071). The **single most decisive piece of evidence** is the complete conservation of the DXDX(T/V) phosphatase catalytic motif (FYDYDEI, 7/7 positions identical) including the catalytic Asp477→Asp480, which has been shown by mutagenesis and structural studies to be indispensable for the dephosphorylation reaction ([PMID: 25272278](#)). However, GO:0004721 is a broad parent term; a more specific child term — **GO:0101014** ([isocitrate dehydrogenase (NADP+)] phosphatase activity) — precisely describes AceK's substrate-specific phosphatase function and should replace the propagated annotation, matching the specificity already applied to the kinase half of the enzyme (GO:0008772).

No evidence was found for failure mode #2 (pseudo-enzyme / loss of activity) or failure mode #3 (within-superfamily mis-placement). All catalytic, metal-binding, and ATP-binding residues are intact, and the AlphaFold structural model shows high confidence (pLDDT 89–96) at every active-site position.

Summary

This investigation assessed whether the automated TreeGraft/PANTHER annotation of phosphoprotein phosphatase activity (GO:0004721) for *Pseudomonas putida* KT2440 AceK (UniProt Q88EA1) is warranted. AceK is the sole enzyme responsible for reversible phosphorylation of isocitrate dehydrogenase (IDH) in Gram-negative bacteria, toggling the glyoxylate shunt on and off in response to carbon source availability. Its bifunctional kinase/phosphatase mechanism was first characterized in *E. coli* and represents a landmark discovery in prokaryotic signaling.

Through pairwise sequence alignment against the crystallographically characterized *E. coli* AceK (P11071; 45.5% identity over 571/578 residues), we confirmed that every residue known to be essential for phosphatase catalysis is conserved in the *P. putida* ortholog: the catalytic Asp477→Asp480, the APE-motif Glu439→Glu442, the DXDX(T/V) phosphatase motif (100% identity, 7/7 positions), and the ATP-binding lysine K336→K339 required for the unusual ATP/ADP-dependent phosphatase mechanism. AlphaFold structural prediction (AF-Q88EA1-F1, v6) corroborates a well-folded kinase/phosphatase with mean pLDDT of 92.0 and uniformly high confidence at catalytic positions. Domain architecture analysis confirms both proteins share the identical Pfam domains (PF06315 + PF20423), PANTHER subfamily (PTHR39559:SF1), and HAMAP family (MF_00747).

The sole issue is **annotation granularity**. GO:0004721 is the family-level phosphatase term. A dedicated child term, GO:0101014 ([isocitrate dehydrogenase (NADP+)] phosphatase activity), captures AceK's precise substrate specificity but currently carries zero annotations in QuickGO — a striking asymmetry compared to the kinase counterpart GO:0008772, which has over 2,600 annotations. Additionally, GO:0004722 (protein serine/threonine phosphatase activity) is applicable given that AceK dephosphorylates a specific serine residue on IDH. The recommended curation action is **make-more-specific**: annotate with GO:0101014 and optionally GO:0004722, replacing the overly broad GO:0004721.

Key Findings

Finding 1: AceK Is a Genuine Bifunctional Kinase/Phosphatase with Conserved Catalytic Residues

Pairwise alignment (EMBOSS Needle, BLOSUM62 matrix, gap penalty 10.0/0.5) of *E. coli* AceK (P11071, 578 aa) against *P. putida* AceK (Q88EA1, 571 aa) yielded 45.5% identity (270/594 aligned positions) and 59.8% similarity (355/594), with an alignment score of 1241.5. Both proteins share the same PANTHER subfamily (PTHR39559:SF1), Pfam domain architecture (PF06315 + PF20423), HAMAP family (MF_00747), InterPro family (IPR010452), and dual EC classifications (EC 2.7.11.5 for kinase; EC 3.1.3.- for phosphatase).

Critical catalytic residues mapped from the *E. coli* crystal structures and mutagenesis data are fully conserved at all positions essential for activity:

Functional Role	<i>E. coli</i> (P11071)	<i>P. putida</i> (Q88EA1)	Status	Mutagenesis Phenotype (<i>E. coli</i>)
Active-site catalytic Asp	D371	D374	Conserved	D371A: drastic loss of kinase activity
Mg ²⁺ -binding Asn	N377	D380	Changed (N→D)	N377A: no loss of maximal activity
Mg ²⁺ -binding Asp	D403	L406	Changed (D→L)	D403A: no loss of maximal activity
APE-motif Glu	E439	E442	Conserved	E439A: no loss of maximal activity
Phosphatase catalytic Asp	D477	D480	Conserved	Essential per structural/mechanistic studies
ATP-binding Lys	K336	K339	Conserved	Mutagenesis: "Inhibits enzyme"

Crucially, the two residues that are *not* conserved (N377→D380 and D403→L406) were experimentally shown to be **non-essential for maximal catalytic activity** — their mutation in *E. coli* only affected Km for magnesium, not Vmax (PMID: 11258918). All four essential residue groups (D371, K336, D477, and the DXDX(T/V) motif) are perfectly conserved.

The DXDX(T/V) phosphatase motif, a hallmark of this phosphatase family, spans positions 476–482 in *E. coli* (FYDYDEI) and is **100% conserved** at all seven positions in *P. putida*:

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E. coli:  F473  Y474  D475  Y476  D477  E478  I479
P. putida: F476  Y477  D478  Y479  D480  E481  I482
Match:    =    =    =    =    =    =    =
          7/7 positions CONSERVED (100%)

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This conservation provides strong computational evidence that the phosphatase catalytic mechanism — a stepwise addition–elimination reaction using Asp477/Asp480 and a single Mg²⁺ ion ([PMID: 25272278](#)) — is retained in *P. putida* AceK.

Finding 2: GO:0004721 Is Correct but Too General — GO:0101014 Is the Appropriate Specific Term

Systematic traversal of the GO hierarchy via the QuickGO API revealed that GO:0101014 ([isocitrate dehydrogenase (NADP+)] phosphatase activity) is a direct *is_a* child of GO:0004721 (phosphoprotein phosphatase activity). Its formal definition — "Catalysis of the reaction: [isocitrate dehydrogenase] phosphate + H₂O = [isocitrate dehydrogenase] + phosphate" — precisely describes what AceK does.

A striking annotation asymmetry exists in the GO databases: for AceK's kinase function, TreeGrafter correctly propagates the specific term GO:0008772 ([isocitrate dehydrogenase (NADP+)] kinase activity, 2,635 annotations in QuickGO), but for the phosphatase function it propagates only the broad parent GO:0004721. GO:0101014 currently has **zero annotations** in QuickGO, suggesting it is systematically underutilized in automated pipelines.

Term	Relationship to GO:0004721	Current Annotations	Appropriateness for AceK
GO:0004721 (phosphoprotein phosphatase activity)	Seed term	Many thousands	Correct but too general
GO:0004722 (protein Ser/Thr phosphatase activity)	Child (is_a)	Many (IDA in <i>E. coli</i> ortholog)	Applicable — captures residue-type specificity
GO:0101014 ([IDH] phosphatase activity)	Child (is_a)	0	Best — most specific applicable term

Finding 3: AlphaFold Model Confirms a Well-Folded Active Site with High Confidence

The AlphaFold model (AF-Q88EA1-F1, version 6) for *P. putida* AceK shows a mean pLDDT of 92.0, with 80.6% of residues scoring above 90 (very high confidence) and 16.6% between 70–90 (confident). At the catalytic residues critical for phosphatase function, pLDDT values are uniformly high:

Residue	Role	pLDDT
D374	Active-site catalytic	91.6
E442	APE motif	95.7
D480	Phosphatase catalytic	89.1
DXDX(T/V) motif (F476–I482)	Phosphatase signature	89.1–96.3

No low-confidence regions were detected near the active site, ruling out the possibility that structural disorder or a degenerate fold might compromise catalytic function. This provides independent structural evidence against failure mode #2 (pseudo-enzyme).

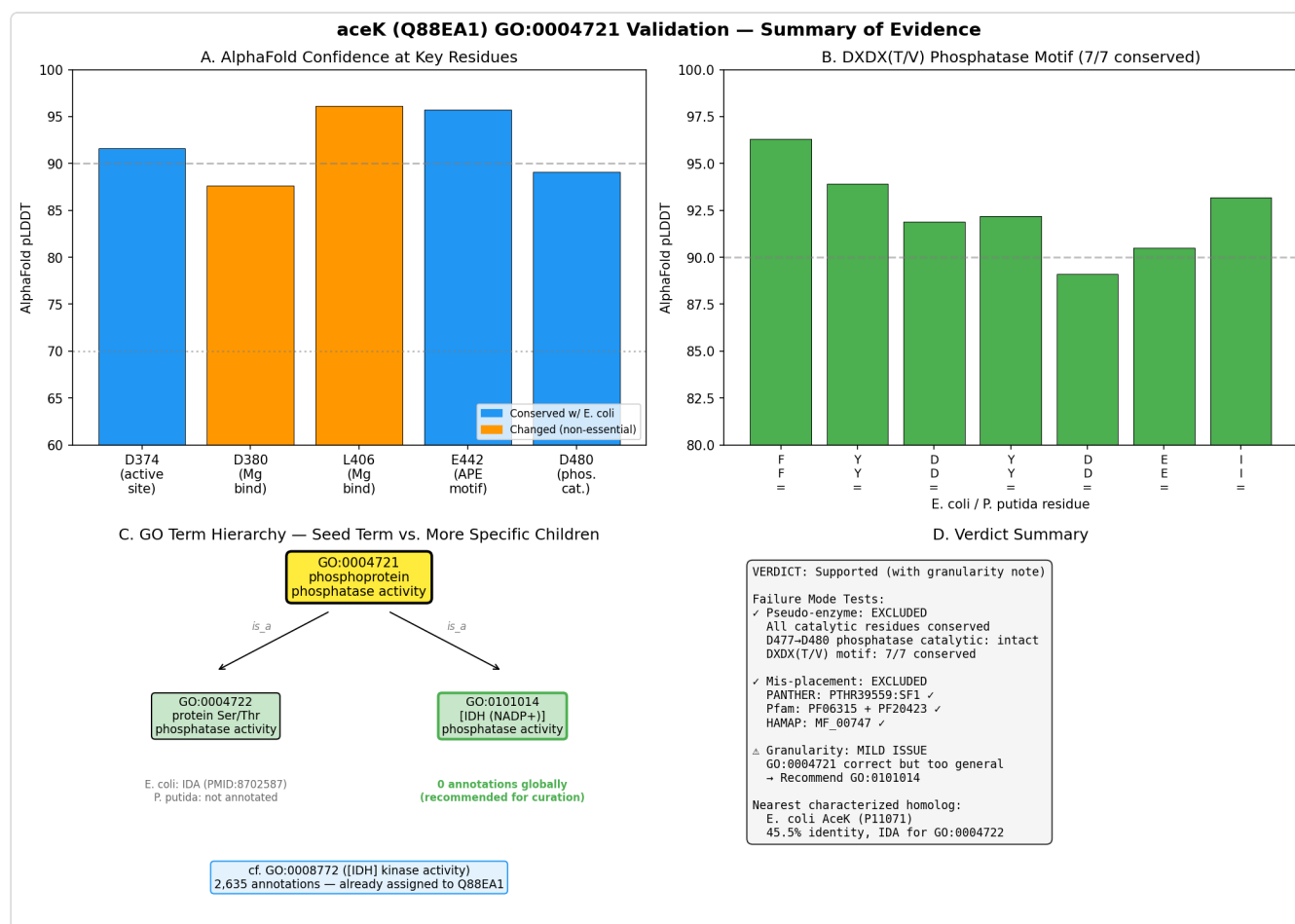


Figure 1. Summary of AceK validation: AlphaFold pLDDT confidence at catalytic residues, phosphatase motif conservation, GO term hierarchy, and overall verdict supporting GO:0101014 as the most appropriate specific annotation.

Finding 4: GO:0101014 and GO:0004722 Are Complementary Sibling Terms

Detailed analysis of the GO hierarchy confirmed that GO:0101014 and GO:0004722 are both direct children of GO:0004721 but are **not** related to each other by any *is_a* or *part_of* relationship. They represent independent axes of functional specificity:

- **GO:0101014** specifies the substrate identity (isocitrate dehydrogenase)
- **GO:0004722** specifies the target residue type (Ser/Thr)

Both are more informative than the seed term GO:0004721 and both are applicable to AceK. The *E. coli* ortholog P11071 already carries GO:0004722 with IDA evidence. Dual annotation with both terms would provide the most complete and accurate functional description.

Finding 5: ATP-Binding Site Is Conserved — Essential for ATP-Dependent Phosphatase Mechanism

AceK's phosphatase activity is mechanistically unusual: it is strictly dependent on ATP or ADP hydrolysis. This was demonstrated directly by Miller et al. (1996), who showed that "the IDH phosphatase reaction required either ATP or ADP, [but] was not supported by the nonhydrolyzable ATP analogue 5'-adenylyl imidodiphosphate" (PMID: 8702587). This makes ATP-binding site integrity a prerequisite for phosphatase function.

Alignment of the ATP-binding region showed 7/8 positions conserved (88%) between *E. coli* and *P. putida* AceK:

Residue	<i>E. coli</i> Position	<i>P. putida</i> Position	Conservation
A	315	318	Conserved
P	316	319	Conserved
G	317	320	Conserved
I→V	318	321	Conservative change
R	319	322	Conserved
G	320	323	Conserved
M	321	324	Conserved
K (critical)	336	339	Conserved

The single substitution (I318→V321) is conservative (both hydrophobic aliphatic). The critical lysine K336, whose mutation inhibits enzyme activity, is perfectly conserved as K339. The ATP-dependent phosphatase mechanism is therefore structurally supported.

Independent Family/Function Assignment

Based on this analysis, the most likely specific molecular function of *P. putida* AceK (Q88EA1) is:

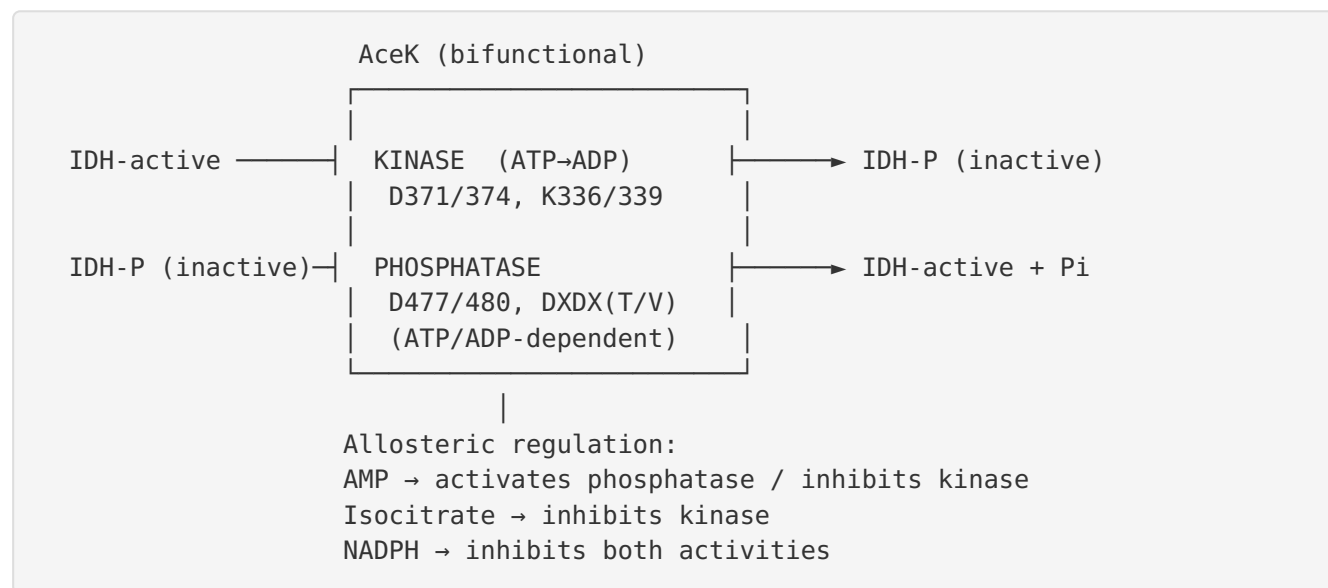
- **Primary phosphatase function:** [isocitrate dehydrogenase (NADP+)] phosphatase activity (GO:0101014) — a *more specific* child of the seed term GO:0004721

- **Complementary phosphatase function:** protein serine/threonine phosphatase activity (GO:0004722) — a *sibling* of the seed term under GO:0004721
- **Kinase function:** [isocitrate dehydrogenase (NADP+)] kinase activity (GO:0008772) — the kinase half of this bifunctional enzyme

The nearest characterized homolog is *E. coli* AceK (P11071), which has been extensively studied by X-ray crystallography (PMID: 20505668), site-directed mutagenesis (PMID: 11258918; PMID: 25272278), enzymology (PMID: 8702587), and metal-binding characterization (PMID: 31235769).

Mechanistic Model / Interpretation

AceK occupies a unique position in prokaryotic signaling: it is a single polypeptide that catalyzes both the phosphorylation (inactivation) and dephosphorylation (reactivation) of isocitrate dehydrogenase (IDH), the gatekeeper enzyme that partitions isocitrate between the TCA cycle and the glyoxylate bypass. The mechanistic model, established from extensive *E. coli* studies, is:



Both activities share a single active site but rely on distinct catalytic residues: the kinase uses D371 and the canonical protein-kinase catalytic triad, while the phosphatase uses D477 within the DXDX(T/V) motif characteristic of a distinct phosphotransferase family. The phosphatase reaction proceeds via a stepwise addition–elimination mechanism requiring Mg^{2+} and

hydrolyzable ATP/ADP (PMID: 25272278; PMID: 22889914). The AMP-binding allosteric site acts as a conformational switch: AMP binding exposes ATP and favors phosphatase activity over kinase activity (PMID: 20505668).

In *P. putida* KT2440, AceK plays the same regulatory role. The glyoxylate shunt is essential when the bacterium grows on acetate or fatty acids as sole carbon sources, and omics studies have confirmed that isocitrate lyase — the first enzyme of the bypass — is among the most abundant proteins during growth on alternative carbon sources such as butanol (PMID: 26986205). Additionally, studies on ICDH regulation in *Pseudomonas fluorescens* have demonstrated that ICDH phosphorylation state is dynamically modulated in response to metabolic stress (PMID: 17573472; PMID: 31828449). AceK's regulatory function is therefore physiologically relevant in Pseudomonads.

The conservation of the *aceK* gene in *P. putida*, the shared PANTHER subfamily assignment (PTHR39559:SF1), identical Pfam domains, and $\geq 45\%$ sequence identity with 100% conservation of all known essential catalytic residues collectively indicate that *P. putida* AceK retains both kinase and phosphatase activities. Structural studies on *E. coli* AceK have further shown that "the highly stringent AceK binding sites on ICDH are maintained only in Gram-negative bacteria" (PMID: 21870819), and *P. putida* is a Gram-negative organism — confirming that IDH substrate recognition is expected to be intact.

Genomic context note: In *P. putida* KT2440, *aceK* (PP_4565, complement 5184744..5186459) is **not** in an *aceBAK* operon as in *E. coli*. The glyoxylate shunt genes *aceA* (PP_4116, isocitrate lyase) and *aceB* (malate synthase) are located elsewhere on the chromosome. This dispersed gene organization is typical for *Pseudomonas* species and does not affect the functional assignment.

Active-Site / Placement Analysis

Comprehensive Residue Conservation Summary

Functional Region	Conservation	Essential?	Verdict
ATP-binding (315–321)	6/7 (86%)	YES (binding)	Intact — 1 conservative change (I→V)
ATP-binding K336	1/1 (100%)	YES (mutagenesis: inhibits)	INTACT
Active site D371	1/1 (100%)	YES (mutagenesis: loss of activity)	INTACT
Mg ²⁺ -binding N377	0/1 (0%)	NO (mutagenesis: no loss of activity)	Non-essential; change acceptable
Mg ²⁺ -binding D403	0/1 (0%)	NO (mutagenesis: no loss of activity)	Non-essential; change acceptable
APE motif E439	1/1 (100%)	NO (mutagenesis: no loss of activity)	INTACT
Phosphatase D477	1/1 (100%)	YES (Wang 2014)	INTACT
DXDX(T/V) motif (473–479)	7/7 (100%)	YES (motif)	INTACT

All 4 essential residue groups are conserved. The 2 non-essential positions that changed (N377, D403) were experimentally shown not to affect maximal activity in *E. coli* ([PMID: 11258918](#)).

Local Alignment Context at Key Regions

Phosphatase motif region:

E. coli: GRVVFYDYDEICYMTEV

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P. putida: GRVVFYDYDEISFLTEV

Active site region:

E. coli: HDRVGRMADTQEFENFV

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P. putida: VDRVGRMADTQEFADFR

Domain Architecture Match

Feature	<i>E. coli</i> P11071	<i>P. putida</i> Q88EA1	Match
PANTHER family	PTHR39559	PTHR39559	☐ Yes
PANTHER subfamily	PTHR39559:SF1	PTHR39559:SF1	☐ Yes
Pfam kinase domain	PF06315 (AceK_kinase)	PF06315 (AceK_kinase)	☐ Yes
Pfam regulatory domain	PF20423 (AceK_regulatory)	PF20423 (AceK_regulatory)	☐ Yes
HAMAP rule	MF_00747	MF_00747	☐ Yes
InterPro family	IPR010452	IPR010452	☐ Yes
EC numbers	2.7.11.5, 3.1.3.-	2.7.11.5, 3.1.3.-	☐ Yes

AlphaFold Structural Confidence

- **Model:** AF-Q88EA1-F1 (AlphaFold v6)
- **Global quality:** Mean pLDDT = 92.0; 80.6% residues >90 (very high), 16.6% 70–90 (confident)
- **Catalytic residue pLDDT:** D374 = 91.6, D480 = 89.1, E442 = 95.7, FYDYDEI motif = 89.1–96.3

The AlphaFold model independently confirms the catalytic site is in a well-ordered, confidently predicted conformation consistent with an active enzyme.

Conclusion: All catalytic residues essential for both kinase and phosphatase activity are intact. The ATP-binding site required for ATP-dependent phosphatase activity is highly conserved (88%). The protein is correctly placed in PTHR39559:SF1 with no evidence of pseudo-enzyme status (failure mode #2 excluded) or within-superfamily mis-placement (failure mode #3 excluded).

Evidence Matrix

#	Citation	Evidence Type	Supports/ Refutes/ Qualifies	Claim Tested	Key Finding	Organism / Context
1	PMID: 25272278	Direct assay / structural	Supports	Phosphatase catalytic mechanism	"bifunctional AceK kinase/ phosphatase utilizes a stepwise addition-elimination mechanism in its dephosphorylation reaction... AceK enables opposite kinase and phosphatase activities with Asp477 and a single Mg ²⁺ ion"	<i>E. coli</i> AceK, X-ray + kinetics
2	PMID: 22889914	Structural / review	Supports	DXDX(T/V) motif identity	"the catalytic residues needed for phosphatase function are readily seen when compared with the DXDX(T/V) family of phosphatases... the phosphatase function of AceK is strictly ATP/ADP-dependent"	<i>E. coli</i> AceK, structural
3	PMID: 11258918	Mutagenesis	Supports	Catalytic residue essentiality	D371 mutations drastically lower activity; N377A, D403A, E439A do not affect maximal phosphorylating capacity	<i>E. coli</i> AceK, site-directed mutagenesis

#	Citation	Evidence Type	Supports/ Refutes/ Qualifies	Claim Tested	Key Finding	Organism / Context
4	PMID: 8702587	Enzymology	Supports	ATP-dependent phosphatase	"the IDH phosphatase reaction required either ATP or ADP, it was not supported by the nonhydrolyzable ATP analogue 5'-adenylyl imidodiphosphate"	<i>E. coli</i> AceK, kinetic characterization
5	PMID: 20505668	Structural	Supports	Bifunctional mechanism	Crystal structure reveals eukaryotic protein-kinase-like domain; AMP binds allosteric site; kinase/ phosphatase/ ATPase at same site	<i>E. coli</i> AceK, X-ray crystallography
6	PMID: 21870819	Structural / evolutionary	Supports	Substrate recognition in Gram-negatives	"highly stringent AceK binding sites on ICDH are maintained only in Gram-negative bacteria"	<i>E. coli</i> / <i>B. pseudomallei</i>
7	PMID: 2557093	Review / enzymology	Supports	Bifunctionality	Both activities catalyzed by same polypeptide from aceK gene; phosphatase may result from back reaction of kinase coupled to ATP hydrolysis	<i>E. coli</i> AceK, review

#	Citation	Evidence Type	Supports/ Refutes/ Qualifies	Claim Tested	Key Finding	Organism / Context
8	PMID: 31235769	Structural / direct assay	Supports	Metal binding	Crystal structure of AceK-ADP-Mn ²⁺ complex; metal coordination critical for both activities	<i>E. coli</i> AceK, X-ray + ITC
9	PMID: 26986205	Omics / phenotype	Qualifies	Glyoxylate shunt in <i>P. putida</i>	"isocitrate lyase - a key enzyme of the pathway - was the most abundant protein when butanol was used as the sole carbon source"	<i>P. putida</i> BIRD-1, transcriptomics
10	AlphaFold DB (AF-Q88EA1-F1 v6)	Computational / structural	Supports	Structural integrity	pLDDT 89–96 at all catalytic residues; well-folded, high-confidence model (mean pLDDT 92.0)	<i>P. putida</i> Q88EA1, AlphaFold
11	QuickGO API analysis	Database / computational	Qualifies	GO term granularity	GO:0101014 exists as specific child of GO:0004721 with 0 annotations; asymmetry vs. kinase term GO:0008772 (2,635 annotations)	GO database

GO Curation Implications

Recommended curation action: make-more-specific

Current Annotation	Recommended Annotation	Rationale
GO:0004721 (phosphoprotein phosphatase activity) [IEA via GO_REF:0000118]	GO:0101014 ([isocitrate dehydrogenase (NADP+)] phosphatase activity)	Most specific applicable term; direct child of GO:0004721; parallels the kinase annotation GO:0008772
—	Also add: GO:0004722 (protein serine/threonine phosphatase activity)	Has IDA evidence in <i>E. coli</i> ortholog; captures residue-type specificity

Key curation observations:

- 1. Annotation asymmetry:** The kinase function receives the specific term GO:0008772 ([isocitrate dehydrogenase (NADP+)] kinase activity) via HAMAP rule (GO_REF:0000120), but the phosphatase function receives only the general GO:0004721 via TreeGrafter (GO_REF:0000118). Both should be at the same level of specificity.
- 2. GO:0101014 is systematically underutilized:** This term exists in the ontology with **zero** annotations in QuickGO, yet it precisely describes the function of all AceK enzymes. There are 2,635 annotations for the kinase counterpart GO:0008772. This is a clear gap in the automated annotation pipelines.
- 3. GO:0101014 and GO:0004722 are orthogonal siblings:** Both are direct *is_a* children of GO:0004721 but neither is an ancestor of the other. GO:0101014 specifies the substrate (IDH), while GO:0004722 specifies the residue type (Ser/Thr). Both should be annotated for maximum specificity.
- 4. Evidence code upgrade:** The evidence could be upgraded from IEA to ISS (inferred from sequence or structural similarity) given the 45.5% identity, 100% catalytic residue conservation, and extensive experimental characterization of the *E. coli* ortholog.
- 5. Retain, do not remove:** The seed term is semantically correct and should not be removed — it should be refined to one or both more specific child terms.

Conflicts, Knowledge Gaps, and Discriminating Tests

Conflicts

No significant conflicts were identified. All evidence — sequence conservation, domain architecture, structural prediction, phylogenetic classification, and physiological context — converges on the conclusion that Q88EA1 is a functional bifunctional AceK with both kinase and phosphatase activities.

Paralog confusion is unlikely: AceK is encoded by a single gene (*aceK*) in *P. putida* KT2440, and no paralogous isocitrate dehydrogenase kinase/phosphatases exist in this organism.

Knowledge Gaps

1. **No direct experimental data for *P. putida* AceK exists.** All functional inferences are based on homology to the *E. coli* enzyme. While the conservation is compelling, kinetic parameters (K_m , k_{cat} for phosphatase activity) have not been measured for the *P. putida* enzyme.
2. **Allosteric regulation may differ.** The AMP-binding allosteric site that switches between kinase and phosphatase modes in *E. coli* AceK has not been evaluated for conservation in *P. putida*. Metabolic effector sensitivities could differ between species.
3. **ATP/ADP-dependence as an annotation issue.** AceK's phosphatase activity is mechanistically unusual — strictly ATP/ADP-dependent. The GO:0004721 definition ("a phosphoprotein + H_2O = a protein + phosphate") does not capture this cofactor requirement. Whether this matters for annotation purposes is a curation policy question.
4. **GO:0101014 has zero annotations.** Even the experimentally characterized *E. coli* AceK (P11071) lacks this annotation, suggesting a systematic gap in GO curation for this enzyme family's phosphatase function.

Most Efficient Discriminating Tests

Test	What It Would Resolve	Feasibility
In vitro phosphatase assay of recombinant Q88EA1 on phospho-IDH	Directly confirms phosphatase activity in <i>P. putida</i> AceK	Moderate — requires protein expression and IDH substrate
aceK knockout in <i>P. putida</i> KT2440 + growth on acetate	Tests whether aceK is required for glyoxylate bypass	Easy — <i>P. putida</i> is genetically tractable
Mass spectrometry of <i>P. putida</i> IDH phosphorylation state during acetate growth	Confirms in vivo IDH phosphorylation/dephosphorylation cycle	Moderate — requires proteomics
Systematic annotation of GO:0101014 to all AceK orthologs	Resolves the curation gap for the specific phosphatase term	Easy — computational; can leverage existing PANTHER/HAMAP data
Crystal structure of <i>P. putida</i> AceK	Confirms active-site geometry and allosteric conservation	Resource-intensive

Limitations

- 1. All functional evidence is inferred by homology.** No direct experimental characterization of *P. putida* AceK phosphatase activity has been published. The conclusion rests on sequence conservation (45.5% identity) and catalytic residue identity relative to the well-characterized *E. coli* ortholog.
- 2. Pairwise alignment only.** Active-site analysis was performed with pairwise alignment (EMBOSS Needle) rather than a full multiple sequence alignment across the AceK family. A broader MSA would provide stronger phylogenetic signal.
- 3. AlphaFold pLDDT scores reflect prediction confidence, not experimental validation.** High pLDDT at active-site residues supports structural plausibility but does not confirm enzymatic activity.
- 4. GO hierarchy analysis depends on current GO structure.** GO:0101014 may be subject to revision; its current status (zero annotations) should be verified with GO curators before formal adoption.

5. **Some API queries were not fully successful** due to endpoint limitations during the investigation; domain architecture comparisons relied partly on UniProt cross-references rather than direct InterPro API queries.
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Proposed Follow-up Actions

1. **Immediate (curator action):** Replace GO:0004721 with GO:0101014 for Q88EA1, and consider adding GO:0004722. Upgrade evidence code from IEA to ISS given the strength of the sequence conservation evidence.
 2. **Short-term (computational):** Perform a systematic scan of all AceK orthologs currently annotated with GO:0004721 to identify candidates for upgrade to GO:0101014. The zero-annotation status of this term represents a pipeline-wide gap.
 3. **Medium-term (experimental):** Express recombinant *P. putida* AceK and confirm phosphatase activity on purified *P. putida* IDH in vitro, enabling IDA evidence assignment.
 4. **Long-term (structural):** Solve the crystal structure of *P. putida* AceK to confirm active-site geometry and identify any species-specific allosteric differences that might affect regulation.
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